

Brian Xu

SENIOR SOFTWARE ENGINEER

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SUMMARY

Innovative software engineer with 7+ years of proven expertise at **Databricks, Google, Reddit and LinkedIn** in designing **robust architectures** and spearheading the development of **highly-scalable** applications and services. Specializing in **backend-heavy** full stack engineering, building and architecting **SaaS Products, distributed systems and data storages & pipelines**. My proficiency spans **Java, Python, Golang, Ruby, Rust** and **JavaScript / TypeScript** and strong proficiency in working with multiple OS including **UNIX, Linux (Ubuntu), MacOS** for server management & development, enabling me to engineer solutions that seamlessly blend technology stacks for maximum efficiency.

SKILLS & ABILITIES

Java

Skilled in building Robust, enterprise-level applications

Golang

Expertise in designing scalable, distributed backend services

Machine Learning

Integrating ML models, LLMs for content generation, chatbot. Document moderation & generation

Python

Experience in Scripting and web services. Data processing, web scraping.

JavaScript & TypeScript

React, Next, Lit, Angular frontend and Node.js backend development expertise with Nest.js, Express.js

Databases

PostgreSQL, MongoDB, MySQL, Neo4j, Cassandra

AWS

ECR, EC2, EKS, S3, DynamoDB, Aurora, CloudFormation

GCP

Firebase, GCloud, GKE, BigQuery, Ads, GTM, GA, AlloyDB

DevOps

GitHub Actions, Gitlab, Jenkins, Terraform, Docker, Kubernetes

EXPERIENCE

Apr 2024 – Present

Law Practice AI (contract @ **KeY2Moon Solutions**) – San Francisco, CA
Senior Software Engineer (Full Stack)

- Led the migration of monolithic backend services built by PHP Laravel to Java Spring Boot microservices, using event-driven architecture with Celery, RabbitMQ and Kafka.
- Developed frontend component libraries using React and Vue 3, upgraded charting from Recharts to Nivo for smoother animations and improved performance.
- Built on-boarding, demo booking and billing systems using Ruby on Rails, ASP.NET Core, React and PostgreSQL.
- Conducted the design and implementation of automated test suites for GraphQL APIs and microservices with JUnit, Mocha, Jest, and Selenium, improving code coverage and quality.
- Developed dynamic pricing and billing systems using Node.js, integrating Stripe APIs for per-tenant usage tracking and automated invoicing in a multi-tenant SaaS platform.
- Implemented API Gateway using Kong API Gateway and Nginx, enhancing security with JWT-based & OAuth2-based authentication, rate-limiting, and traffic routing.

- Developed AI Services for virtual assistant chatbot utilizing LLM (ChatGPT, Gemini) with integration of RAG pipeline, as well as OCR models for legal document process and implemented prompt engineering practices for cost effective model moderation.
- Developed RESTful APIs and GraphQL services using Spring Boot, Go, and integrated them with third-party services built in Python (Flask) and Node.js (Express.js & Nest.js).
- Refactored legacy ELT data pipeline by migrating to Snowflake and Apache Airflow, improving performance, scalability, and reducing processing time by 40%.
- Orchestrated multi-agent service for automated legal document moderator and generator utilizing LLM, NLP and OCR models with AWS SageMaker and LangChain, LangGraph.
- Managed high-availability cloud deployments on AWS EC2, leveraging Auto Scaling, Load Balancing, and Spring Boot Actuator for system monitoring and resilience.

Nov 2022 – Apr 2024

Databricks (*contract @ Key2Moon Solutions*) – Mission, TX

Senior Software Engineer (Backend)

- Designed cloud-agnostic infrastructure using Kubernetes and Terraform to power big data and ML workloads on Databricks, enabling scalability across AWS, Azure and GCP.
- Architected multi-tenant SaaS infrastructure in AWS, ensuring secure data isolation and scalability for enterprise customers.
- Collaborated closely with front-end teams to achieve seamless integration using JavaScript & TypeScript (Next.js, Vue.js and React), prioritizing user-centric services and integrated with Firebase (GCP).
- Built resource management Golang, Ruby-based services for distributed system, optimizing compute and storage for Apache Spark and Delta Lake, reducing costs by 30%.
- Pioneered the integration of Salesforce CRM with SaaS enterprise applications, contributing to a 20% increase in operational efficiency.
- Developed microservices in Python (Django, FastAPI) and Java Spring Boot, automating cloud-specific configurations and simplifying multi-cloud operations for Databricks engineers.
- Migrated legacy applications to Ruby on Rails, reducing technical debt and improving maintainability, adopting TDD with RSpec for high-quality, bug-free code.
- Optimized job scheduling and orchestration systems with Apache Airflow and Java for large-scale data pipelines, reducing processing time in cross-region deployments.
- Streamlined integration of machine learning models with distributed data processing, ETL pipeline enabling real-time model training and inference at scale with MLflow.
- Built fault-tolerant distributed systems with Spring Cloud, Kubernetes, and Kafka, ensuring geo-redundancy and high availability for ML and big data applications.
- Led internal tooling efforts with Docker and K8S to enhance Databricks' cloud portability, increasing engineer productivity and reducing deployment errors.
- Contributed to a project to migrate from a traditional RDBMS to Neo4j, which reduced data redundancy and improved data integrity while handling over 2 million nodes and relationships.
- Implemented MongoDB Atlas with Spring Data MongoDB, reducing infrastructure costs while ensuring high performance and security.
- Optimized centralized monitoring with Spring Boot Actuator, Prometheus, Grafana, and ELK Stack, improving observability and log analytics.
- Enhanced distributed caching strategies using Spring Cache with Redis, optimizing database interactions for both SQL (PostgreSQL) and NoSQL (MongoDB, Neo4j) to improve system responsiveness.

Apr 2021 – Nov 2022

Reddit (contract @ **KeY2Moon Solutions**) – Mission, TX
Senior Software Engineer (Full Stack)

- Built and optimized Go, Java and Python microservices for post editing, voting and reporting through optimized DB SQL queries and migration of REST to gRPC services.
- Conducted the migration of the content database from a monolithic PostgreSQL schema to Cassandra clusters, improving write throughput by 2x for comments and posts.
- Consolidated the event-driven framework to support real-time updates for post/comment interactions, integrating WebRTC as well as Kafka and RabbitMQ to ensure reliable message delivery.
- Authored backend APIs and documentation using OpenAPI (Swagger), improving post/comment debugging and reducing incident response times for engineering teams.
- Implemented complex search features for Reddit posts and comments using Elasticsearch allowing users to query historical content with full-text search and filtering capabilities.
- Maintained and updated a GraphQL layer to allow retrieval and updates of user interactions with PostgreSQL database.
- Contributed to the enhancement of frontend and API integration part of post/comment UI by utilizing React, Lit as frontend libraries, and Go/Gin as backend service.
- Developed complex web applications using Go (Golang), JavaScript/TypeScript and Java Spring Boot, working closely with UI/UX teams to deliver superior user experiences.
- Introduced auto-scaling strategies in Kubernetes to dynamically handle surges in traffic, maintaining service availability during unpredictable demand spikes.
- Improved service throughput by 30% by optimizing batch processing and data pipelines using Apache Kafka, Flink for efficient message streaming and stream processing.
- Collaborated with other development teams to integrate application-level authentication with AWS IAM, leveraging Amazon Cognito for secure and scalable user authentication.

Oct 2018 – Apr 2021

Google (full-time) – Mountain View, CA
Software Engineer (Full Stack)

- Led the migration of security services to a microservices-based architecture with Go, Java and Python frameworks, improving scalability and fault tolerance, supporting traffic spikes.
- Worked on Google's internal encryption framework, ensuring sensitive user data is securely stored in compliance with AES-256, TLS 1.3 industry-leading standards.
- Optimized security protocols for data encryption at rest and in transit, ensuring compliance with global privacy regulations including GDPR, CCPA for over 1B Google users.
- Strengthen the real-time anomaly detection system to identify security threats and unauthorized access patterns by integrating ML models on user activity logs and traffic data.
- Collaborated with cross-functional teams to integrate security measures into the distributed data layer, enforcing data encryption and implementing role-based access control (RBAC).
- Managed data consistency across distributed databases, utilizing strong and eventual consistency models to ensure data integrity and availability in multi-region environments.
- Implemented graceful service degradation strategies for mission-critical services, using circuit breakers with Hystrix and rate limiting to maintain operational stability.
- Architected multi-region database replication strategies, leveraging Google Cloud Spanner and Cassandra for geo-replication, ensuring minimal latency and zero data loss.
- Developed a global distributed caching mechanism using Redis and Google Cloud CDN, enabling low-latency access to cached content while reducing API response times by 12%.
- Built fault-tolerant backend systems with auto-scaling capabilities with GKE and Google Cloud Load Balancer, ensuring seamless performance during spikes in global traffic.

- Implemented backup and disaster recovery strategies for Cassandra data stores, significantly reducing the risk of data loss.
- Built RESTful APIs using Gin-Gonic | Go Echo | Gorilla-Mux packages in Go (Golang) and used mongo-driver and PQ/SQL packages for accessing NoSQL and SQL database.
- Engineered a distributed data pipeline using Go, Python and PostgreSQL, synchronizing telemetry data for real-time decision-making, increasing data reliability.
- Played pivotal role in enhancing streaming services using gRPC and protobuf in Golang and Ruby-based microservice architecture especially in event-driven architecture.
- Designed and maintained OpenAPI specs for RESTful microservices, ensuring clear API contracts, versioning, and seamless integration across teams.

May 2017 – Aug 2017

LinkedIn (*intern*) – Sunnyvale, CA
Software Engineer Intern

- Implemented Python scripts and Java modules to automate tasks, reducing manual work by 30% and optimized ensuring they adhered to SRE best practices.
- Participated in incident management and post-mortem reviews, identifying key bottlenecks in system performance and contributing to the implementation of performance optimizations that reduced latency by 15%.
- Contributed to complex data analysis initiatives, executing SQL queries and PL/SQL stored procedures monthly, improved data integrity and reporting mechanisms.
- Built the backend part of internal tools for development management of the company using Express.js in Node.js, microframeworks and routing packages in Go (Golang) and PostgreSQL.

EDUCATION

Massachusetts Institute of Technology

Sep 2014 – May 2018

Bachelor's Degree – Computer Science

Sep 2017 – Jun 2018

Master of Engineering – Computer Science

INTERESTS & SELF-DEVELOPMENT

Strong interest in Rust, Blockchain, smart contracts, and Web3 technologies for cutting-edge modern web applications. Actively engaging in AI/ML world, with a focus on building and managing real-world chat systems and generative AI applications.